Unit 3: The Precise Definition of a Limit — Full Solutions

Warm-up

1. $f(x) = 2x$, $a = 3$, $L = 6$, $\varepsilon = 0.4$.

Here the slope is $m = 2$. Using the shortcut $\delta = \dfrac{\varepsilon}{|m|}$:

$$\delta = \frac{0.4}{2} = 0.2$$

Answer: $\delta = 0.2$. (As long as x stays within 0.2 of 3, $f(x)$ stays within 0.4 of 6.)

2. $f(x) = 5x$, $a = 2$, $L = 10$, $\varepsilon = 1$.

Slope $m = 5$.

$$\delta = \frac{1}{5} = 0.2$$

Answer: $\delta = 0.2$.

3. $f(x) = -3x$, $a = 1$, $L = -3$, $\varepsilon = 0.6$.

Slope $m = -3$, so $|m| = 3$ (the sign doesn't matter here, only the size).

$$\delta = \frac{0.6}{3} = 0.2$$

Answer: $\delta = 0.2$.

4. $f(x) = x + 4$, $a = 5$, $L = 9$, $\varepsilon = 0.3$.

Slope $m = 1$.

$$\delta = \frac{0.3}{1} = 0.3$$

Answer: $\delta = 0.3$. (When the slope is exactly 1, delta and epsilon always match.)

5. Which line needs the smaller delta if both have the same epsilon, but one is steeper?

The steeper line (larger $|m|$) needs the smaller delta. Here's why: a steep line takes a small change in x and stretches it into a much bigger change in y — think of it like a lever that amplifies a small push into a big swing. So to keep the output change small (within ε), you have to control the input change even more tightly. That's exactly why $\delta = \varepsilon/|m|$ has $|m|$ on the bottom: a bigger $|m|$ makes δ smaller.

6. If someone tightens epsilon, what happens to delta?

Delta must also get smaller (or stay the same at most). Since $\delta = \varepsilon/|m|$, shrinking ε directly shrinks δ (for a fixed slope). This makes sense with the tolerance-game picture: if someone demands a narrower target band on the output, you have to squeeze your input window tighter to still guarantee landing inside it.

Standard

7. $f(x) = 4x - 1$, $a = 2$, $L = 7$, $\varepsilon = 0.8$.

Slope $m = 4$.

$$\delta = \frac{0.8}{4} = 0.2$$

Answer: $\delta = 0.2$.

8. $f(x) = -2x + 5$, $a = 3$, $L = -1$, $\varepsilon = 0.5$.

Slope $m = -2$, so $|m| = 2$.

$$\delta = \frac{0.5}{2} = 0.25$$

Answer: $\delta = 0.25$.

9. $f(x) = 6x + 2$, $a = -1$, $L = -4$, $\varepsilon = 1.2$.

Slope $m = 6$.

$$\delta = \frac{1.2}{6} = 0.2$$

Answer: $\delta = 0.2$.

10. $f(x) = 3x$, $a = 0$, $\varepsilon = 0.09$.

Slope $m = 3$.

$$\delta = \frac{0.09}{3} = 0.03$$

Answer: $\delta = 0.03$.

11. $f(x) = 2x + 1$, $a = 4$, $\delta = 0.15$. Find the resulting ε.

Here we work backward using $\varepsilon = |m| \cdot \delta$, with slope $m = 2$:

$$\varepsilon = 2 \times 0.15 = 0.3$$

First find L: $L = f(4) = 2(4) + 1 = 9$.

Answer: $\varepsilon = 0.3$. In words: whenever x stays within 0.15 of 4, $f(x)$ is guaranteed to stay within 0.3 of 9 — that is, $f(x)$ stays strictly between 8.7 and 9.3.

12. $f(x) = -4x + 3$, $a = 1$, $\delta = 0.1$. Find the resulting ε.

Slope $m = -4$, so $|m| = 4$.

$$\varepsilon = 4 \times 0.1 = 0.4$$

Find L: $L = f(1) = -4(1) + 3 = -1$.

Answer: $\varepsilon = 0.4$. In words: whenever x stays within 0.1 of 1, $f(x)$ is guaranteed to stay within 0.4 of -1 — that is, $f(x)$ stays strictly between -1.4 and -0.6.

Challenge

13. $f(x) = 7x - 3$, $a = 2$, $\varepsilon = 0.35$.

Slope $m = 7$.

$$\delta = \frac{0.35}{7} = 0.05$$

Find L: $L = f(2) = 14 - 3 = 11$.

Check at $x = a + \delta = 2.05$:

$$f(2.05) = 7(2.05) - 3 = 14.35 - 3 = 11.35$$

$$|11.35 - 11| = 0.35 \quad \checkmark \text{ (right at the edge of the band, as expected)}$$

Check at $x = a - \delta = 1.95$:

$$f(1.95) = 7(1.95) - 3 = 13.65 - 3 = 10.65$$

$$|10.65 - 11| = 0.35 \quad \checkmark \text{ (also right at the edge)}$$

Answer: $\delta = 0.05$, and both boundary checks confirm $f(x)$ lands exactly on the edge of the ± 0.35 tolerance band around $L = 11$ — any x strictly inside the δ-window gives an $f(x)$ strictly inside the band, exactly as promised.

14. $f(x) = 3x + 1$, $a = 2$, $L = 7$, $\varepsilon = 0.6$, with an extra requirement that x stay within 0.15 of a.

First, the shortcut formula's delta: slope $m = 3$,

$$\delta_{\text{formula}} = \frac{0.6}{3} = 0.2$$

But we're also told x must stay within 0.15 of $a = 2$ for a separate reason. We have two requirements:

- Requirement 1 (epsilon-matching): $\delta \leq 0.2$
- Requirement 2 (physical constraint): $\delta \leq 0.15$

Answer: use $\delta = 0.15$, the smaller of the two. This is because whatever delta you pick has to satisfy both requirements at once — picking the larger one (0.2) would violate the physical constraint. When you have more than one condition to satisfy, you always take the tightest (smallest) delta among them, since a smaller delta automatically still satisfies any looser requirement.

15. $f(x) = -5x+2$, $a = -2$, $\varepsilon = 0.25$.

First find L: $L = f(-2) = -5(-2)+2 = 10+2 = 12$.

Slope $m = -5$, so $|m| = 5$.

$$\delta = \frac{0.25}{5} = 0.05$$

Is $\delta = 0.02$ also valid?

Yes. Since $0.02 < 0.05$, using $\delta = 0.02$ means we're restricting x to an even tighter window around $a = -2$ than we strictly needed to. A tighter input window can only make $f(x)$ land even closer to L, never farther — so any delta smaller than the one given by the shortcut formula still works. There's no single "correct" delta; 0.05 is just the largest one that's guaranteed to work.

16. $f(x) = 2x$ near $a = 3$, with $f(x)$ required to stay strictly between 5.7 and 6.3.

First, find L: since the band is symmetric, L is the midpoint:

$$L = \frac{5.7+6.3}{2} = \frac{12}{2} = 6$$

(This matches $f(3) = 2(3) = 6$, as expected.)

Find ε: it's the distance from L to either edge:

$$\varepsilon = 6.3-6 = 0.3 \quad \text{(or equivalently } 6-5.7 = 0.3\text{)}$$

Now find the matching δ, using slope $m = 2$:

$$\delta = \frac{0.3}{2} = 0.15$$

Answer: $\varepsilon = 0.3$ and $\delta = 0.15$.

Applied

17. Rod-cutting: $f(x) = 2x$, target $L = 6$ m at $a = 3$, tolerance $\varepsilon = 0.02$ m.

Slope $m = 2$.

$$\delta = \frac{0.02}{2} = 0.01$$

Answer: the dial must be set within 0.01 units of 3 to guarantee the rod comes out within 0.02 m of the 6 m target.

18. Thermostat: $T(x) = 3x + 65$, target $L = 71°F$ at $a = 2$, tolerance $\varepsilon = 0.9°F$.

Slope $m = 3$.

$$\delta = \frac{0.9}{3} = 0.3$$

Answer: the dial must be set within 0.3 units of 2 to guarantee the room stays within $0.9°F$ of $71°F$.

19. Pharmacist: $C(x) = 4x + 10$, target $L = 26$ mg/mL at $a = 4$, tolerance $\varepsilon = 0.5$ mg/mL.

Slope $m = 4$.

$$\delta = \frac{0.5}{4} = 0.125$$

Answer: x must be measured within 0.125 mL of 4 mL to guarantee the concentration stays within 0.5 mg/mL of 26 mg/mL.

20. Drone: $h(x) = 10x$, target $L = 50$ m at $a = 5$, tolerance $\varepsilon = 2$ m.

Slope $m = 10$.

$$\delta = \frac{2}{10} = 0.2$$

Answer: the throttle setting must be controlled within 0.2 units of 5 to guarantee the altitude stays within 2 m of 50 m, keeping the drone clear of the restricted zone.